1 Fig. 1.1 shows a coil (coil X).

For Examiner's Use

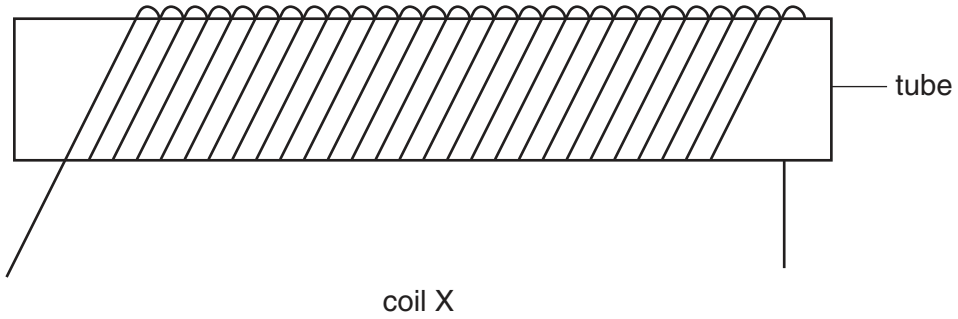

Fig. 1.1

A student winds another coil (coil Y) tightly around coil X.

A changing e.m.f. in coil X induces an e.m.f. in coil Y.

The student wishes to investigate how the e.m.f. V in coil Y depends on the frequency f of the current in coil X.

It is suggested that V is directly proportional to f.

Design a laboratory experiment to investigate the suggested relationship. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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